

ECON 470 Homework 5

Submission 3, Spring 2026

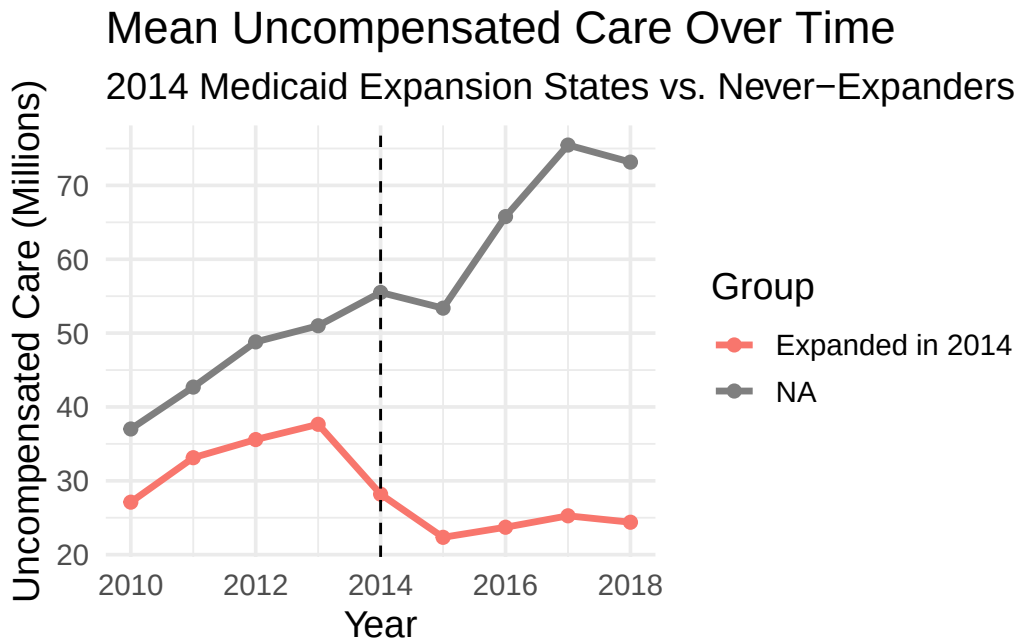
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Problem 1

Table 1: Uncompensated Care (\$M), 2010–2018

Year	Mean (\$M)	Std. Dev. (\$M)
2010	30.26	72.26
2011	34.10	74.96
2012	37.54	86.63
2013	39.43	80.86
2014	36.63	88.53
2015	33.37	86.90
2016	45.24	403.12
2017	41.14	102.67
2018	39.10	100.40

Problem 2



Yes, the graph does suggest a potential treatment effect. We note that the states with adoption in 2014 have decreased their amount of uncompensated care as opposed to states that are never-expanders or yet-to-expanders.

Problem 3

Table 2: Difference-in-Differences: Mean Uncompensated Care (\$M)

Group	2012 (\$M)	2015 (\$M)	Difference (\$M)
Control	48.81	53.38	4.57
Treatment	35.58	22.35	-13.24

Problem 4

From the 2 by 2 DiD table, we note that states that might have adopted the Medicaid expansion policy in 2014 have decreased their amount of uncompensated care from 2012 to 2015, whereas never-expanders (or yet-to-expanders) in the control group have rather faced increased amounts of uncompensated care as observed. Thus, the adoption of Medicaid leads to a decrease in uncompensated care. Further, we expect this effect as it encourages hospitals to take Medicare patients and get compensation or partial support from the Medicare program, instead of outsourcing debt to manage or counter their listed charges to third parties, loaning ventures, or debt collector banks.

Estimate ATEs

Problem 5

Table 3: DiD Estimates: Effect of Medicaid Expansion on Uncompensated Care (\$M)

Term	Estimate	Std. Error	t-Statistic	p-Value
Intercept	47.976	2.104	22.799	3.87e-113 ***
Post 2014	15.938	2.677	5.953	2.70e-09 ***
Ever Expanded	-12.419	2.653	-4.682	2.87e-06 ***
Post x Ever Expanded (DiD)	-26.681	3.382	-7.888	3.27e-15 ***

Observations: 14990

R-squared: 0.024

Signif. codes: *** p<0.01, ** p<0.05, * p<0.10

Problem 6

Table 4: TWFE Estimates: Effect of Medicaid Expansion on Uncompensated Care (\$M)

Term	Estimate	Std. Error	t-Statistic	p-Value
Treatment (DiD)	-30.755	5.903	-5.21	7.36e-06 ***

Hospital and year fixed effects included.

Standard errors clustered at the state level.

Observations: 14990

R-squared (within): 0.039

Signif. codes: *** p<0.01, ** p<0.05, * p<0.10

We notice that both the estimators of Difference in Differences and Two-Way Fixed Effects yield near about same magnitude of estimated effect, around 26 to 30 million of uncompensated care in absolute magnitude, with the same number of observations and a statistically significant p-value.

Problem 7

Table 5: OLS DiD: Effect of Medicaid Expansion on Uncompensated Care (\$M), All States

Term	Estimate	Std. Error	t-Statistic	p-Value
Intercept	47.976	3.643	13.171	1.88e-39 ***
Post 2014	15.938	4.635	3.439	5.85e-04 ***
Ever Expanded	-14.916	4.292	-3.475	5.11e-04 ***
Post x Ever Expanded (DiD)	-19.504	5.461	-3.572	3.56e-04 ***

Observations: 20095

R-squared: 0.006

Signif. codes: *** p<0.01, ** p<0.05, * p<0.10

Table 6: TWFE: Effect of Medicaid Expansion on Uncompensated Care (\$M), All States

Term	Estimate	Std. Error	t-Statistic	p-Value
Treatment (DiD)	-23.609	4.736	-4.985	7.54e-06 ***

Hospital and year fixed effects included.

Standard errors clustered at the state level.

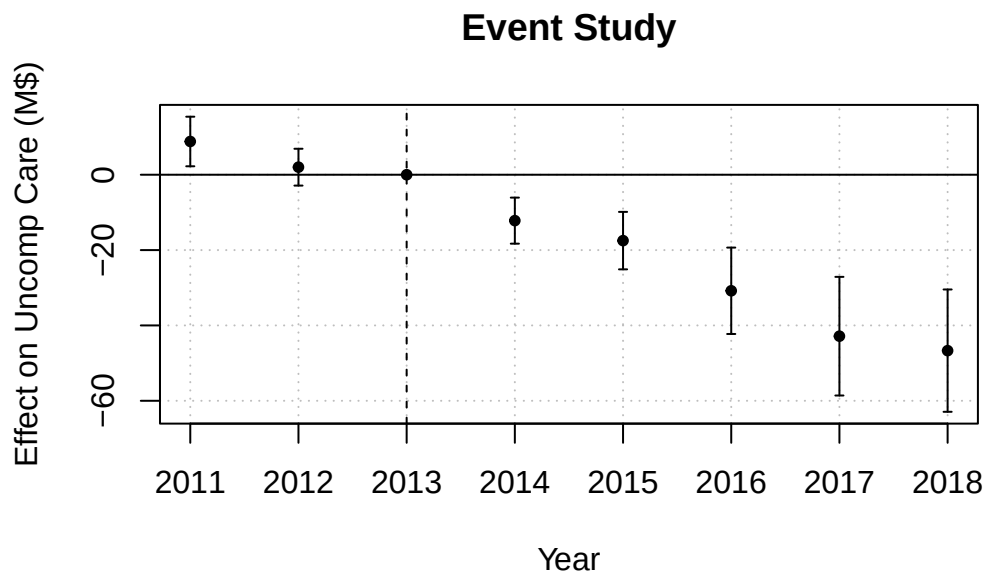
Observations: 20095

R-squared (within): 0.002

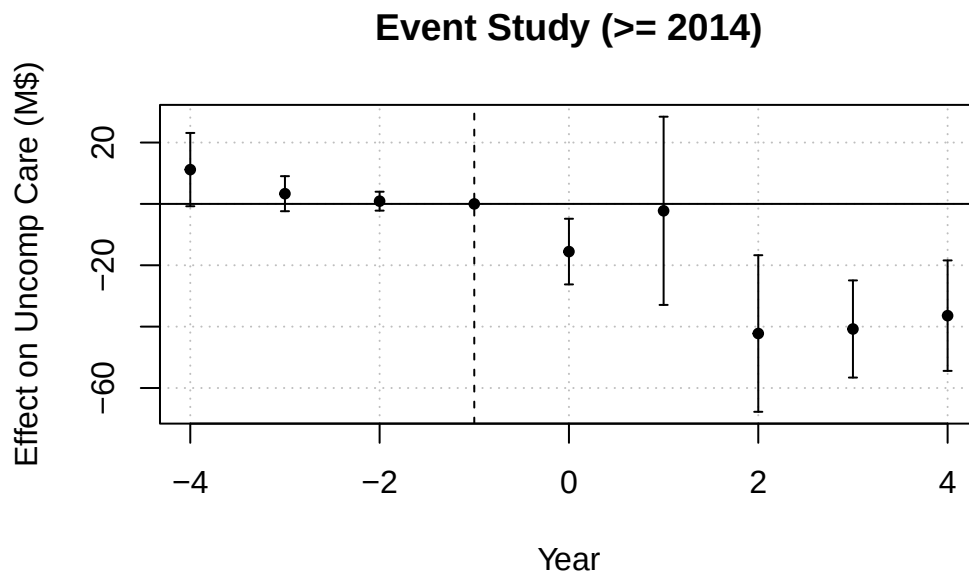
Signif. codes: *** p<0.01, ** p<0.05, * p<0.10

Yes, the results vary slightly when we include all states (even those that expanded after 2014). We notice a lower absolute value of uncompensated care since this cumulatively aggregates all the treatment effects across the different states, including those that might have (and would have) different adoption years, or might not have expanded ever. This doesn't take into account that the number of control years could then vary according to the year of adoption, and similarly, the years of post-adoption would be different too. Further, unvieling TWFE as a biased estimator.

Problem 8



Problem 9



Problem 10

Medicaid expansion leads to a decrease in uncompensated care for hospitals in states that have adopted the change in policy. Event study graphs are concerning when the bars for pre-treatment deviate considerably far from the zero margin, showing a violation of the parallel trends assumption. However, in our case, we see the confidence interval of our pre-treatment variates to include 0 and further congregate around zero margin on average, following which, when treatment comes into effect, we do see a significant effect of a downward slope. The invocation of the event time as opposed to a simple calendar year takes care of the previous bias of different states having different adoption years, and further portrays and brings a more centralized angle to this analysis as opposed to the primary limitation of the TWFE estimator being biased in nature, particularly because of lacking this staggered time context. Furthermore, the binning effect around the ends accounts for aggregating the effect of states that might have accumulatively expanded over a certain period but not over another, so this spreads out the congregated effect and accurately captures the right treatment effect, being more granular instead of biasing towards particular years and states.